TEAM EAGLE SHIELD

Team 25

University of Tirana, Faculty of Economy

WHITE TEAM DOCUMENTATION

International Cybersecurity Exercise (ICE) 2026

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1. Public Servers Overview

The following table lists all public-facing servers assigned to Team 27 during the ICE 2024 competition.

Domain	IP Address	Ports
admin.team25.isucdc.com	108.64.203.40	80, 3389
tickets.team25.isucdc.com	108.64.203.20	3389, 5000
mgmt.team25.isucdc.com	108.64.203.30	22, 5000
ad.team25.isucdc.com	108.64.203.10	389,636,3389
www.team25.isucdc.com	108.64.203.50	80, 22

2. Server Documentation

2.1 Active Directory (AD)

- **Operating System**: Windows Server 2016
- **Domain:** ad.team25.isucdc.com
- **IP Address:** [Assign your team IP]
- **Description:**
  The Active Directory server manages user authentication for all machines and website logins. It also provides certificates required by the Ticketing Machine. All scenario machines must be joined to this AD server.
- **Configuration Steps:**
  - Verified all servers joined to the AD domain
  - Enabled LDAP on TCP port 389 and LDAPS on TCP port 636
  - Configured RDP on TCP port 3389 for IT Admin access
  - Tested certificate requests from the Ticketing Machine using certutil -config - -ping
- **Required Access:**
  - LDAP: port 389

  - LDAPS: port 636
  - RDP: port 3389 (Admin access)
  - Ticketing Machine certificate requests
- **Flags:**
  - Red: C:\Users\Administrator
  - Blue: C:\Windows\System32

2.2 Ticketing Machine

- **Operating System:** Windows Server 2019
- **Domain:** tickets.team25.isucdc.com
- **IP Address:** 108.64.203.20
- **Description:**
  Hosts the application for issuing and validating tickets for buses and trains. Represents the city's backend ticketing system and physical ticket readers.
- **Configuration Steps:**

  Navigated to C:\Users\Administrator\Desktop\tickets

  Compiled and deployed ticketing application according to README

  Verified application runs on boot via scheduled task

  Configured HTTP access on port 5000

  Configured RDP on port 3389
- **Required Access:**

  RDP: port 3389 (Admin & Ticket Maintenance)

  HTTP: port 5000 (Backend and competition network)

- **Flags:**

  Red: C:\Users\Administrator

  Blue: C:\Windows\System32

  Forge Ticket: validUntil ≥ 2000000000

2.3 Management Backend

- **Operating System**: AlmaLinux 8.4
- **Domain**: mgmt.team25.isucdc.com
- **IP Address**: 108.64.203.30
- **Description**:
  Main transit management backend for authentication, ticket payments, and fleet/route management. Connects to Admin UI, Public Website, and simulated vehicles.
- **Configuration Steps:**

  Navigated to /home/cdc/icdc26-backend-mgmt

  Configured backend systemd service to start on boot

  Verified SSH access on port 22 and HTTP on port 5000

  Configured communication with simulated buses/trains
- **Required Access:**

  SSH: port 22 (Admin)

  HTTP: port 5000 (Admin UI, Public Website)
- **Flags:**

  Red: /root/

  Blue: /etc/

Database Access: retrieve blue_flag payment record

Service Disruption: /vehicles/<bus or train>/flag

2.4 Admin User Interface

- **Operating System**: Windows Server 2022
- **Domain**: admin.team25.isucdc.com
- **IP Address:** 108.64.203.40
- **Description:**
  Admin portal for viewing city transportation status and managing fleet.
- **Configuration Steps:**

  Navigated to C:\Users\Administrator\Desktop\icdc26-admin-ui

  Verified scheduled task “Admin UI” runs on boot

  Configured HTTP access on port 80

  Configured RDP access on port 3389
- **Required Access:**

  HTTP: port 80 (Admin features)

  RDP: port 3389 (Admin access)
- **Flags:**

  Red: C:\Users\Administrator

  Blue: C:\Windows\System32

  Service Disruption: disable buses/trains via portal button

2.5 Public Website

- **Operating System:** AlmaLinux 9
- **Domain:** www.team25.isucdc.com
- **IP Address:** 108.64.203.50
- **Description:**
  Hosts the city's public website for ticket purchase, account management, and scanning tickets.
- **Configuration Steps:**

  Navigated to /home/cdc/icdc2026frontend-main

  Configured frontend systemd service to run on boot

  Enabled HTTP access on port 80

  Verified SSH administrative access on port 22
- **Required Access:**

  HTTP: port 80 (public and admin users)

  SSH: port 22 (Admin access)
- **Flags:**

  Red: /root/

  Blue: /etc/

3. SCADA Infrastructure

Our team designed and implemented a SCADA infrastructure to oversee and manage the public transportation system in Central Downtown City (CDC). Leveraging a combination of software and networked hardware components, our SCADA system serves as the central nervous system of the transportation network, enabling real-time monitoring, control, and analysis of critical parameters.

Architecture

At the core of our SCADA infrastructure lies the **Management Backend server**, meticulously configured to facilitate seamless communication between buses, trains, the Frontend web application, and the ticketing system. This server acts as the backbone of the system, logging essential data such as vehicle status, ticket transactions, and system health, while also serving as the intermediary for data exchange between various components.

Communication

Our SCADA system incorporates **MQTT (Message Queuing Telemetry Transport)** technology, implemented through the **MQTT Broker**, to ensure efficient and reliable communication between the vehicles' simulation machines and the Management Backend server. The Frontend and Admin web applications provide user-friendly interfaces for operators to monitor vehicle status, manage ticketing, view camera feeds, and control aspects of the transportation network in real-time.

Security

To bolster security and integrity, our SCADA infrastructure is tightly integrated with the **Active Directory (AD) server**, ensuring robust user authentication and access control measures. Regular monitoring protocols are established to promptly identify and mitigate potential vulnerabilities, service disruptions, or unauthorized access attempts.